

Limits of Schmidt Decomposition and Tripartite Entanglement

Problem Statement

One might hope that a Schmidt decomposition also exists for tripartite states, i.e. that any three-party pure state could be written as

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle |i_B\rangle |i_C\rangle,$$

with $\lambda_i \geq 0$ and orthonormal bases $\{|i_A\rangle\}$, $\{|i_B\rangle\}$, and $\{|i_C\rangle\}$.

(a) Assuming $|\psi\rangle$ is of this form, compute the spectra of

$$\rho^A = \text{tr}_{BC} |\psi\rangle\langle\psi|, \quad \rho^B = \text{tr}_{AC} |\psi\rangle\langle\psi|, \quad \rho^C = \text{tr}_{AB} |\psi\rangle\langle\psi|.$$

(b) Using (a), show that

$$|\phi\rangle = \frac{1}{2} |000\rangle + \frac{1}{2} |011\rangle + \frac{1}{\sqrt{2}} |100\rangle$$

cannot be written in Schmidt form.

(c) Consider the states

$$|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle), \quad |\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle).$$

Compute $\text{tr}_3 |W\rangle\langle W|$ and $\text{tr}_3 |\text{GHZ}\rangle\langle \text{GHZ}|$ and test entanglement using the Peres–Horodecki (PPT) criterion.

(a) Reduced Density Matrices

Assume

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle |i_B\rangle |i_C\rangle.$$

Then

$$\rho^A = \text{tr}_{BC} |\psi\rangle\langle\psi| = \sum_i \lambda_i^2 |i_A\rangle\langle i_A|.$$

Similarly,

$$\rho^B = \sum_i \lambda_i^2 |i_B\rangle\langle i_B|, \quad \rho^C = \sum_i \lambda_i^2 |i_C\rangle\langle i_C|.$$

Key Result: All three reduced density matrices have the *same spectrum*:

$$\boxed{\text{Spec}(\rho^A) = \text{Spec}(\rho^B) = \text{Spec}(\rho^C) = \{\lambda_i^2\}.$$

(b) No Schmidt Decomposition Exists

For

$$|\phi\rangle = \frac{1}{2} |000\rangle + \frac{1}{2} |011\rangle + \frac{1}{\sqrt{2}} |100\rangle,$$

compute reduced states.

Tracing over BC :

$$\rho^A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \text{eigenvalues } \{1/2, 1/2\}.$$

Tracing over AC :

$$\rho^B = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}, \quad \text{eigenvalues } \{3/4, 1/4\}.$$

Since ρ^A and ρ^B have *different spectra*, this contradicts the necessary condition from part (a).

$$\boxed{|\phi\rangle \text{ cannot be written in Schmidt form.}}$$

(c) Robustness of Entanglement

W State

Tracing out qubit 3:

$$\rho_{AB}^{(W)} = \frac{1}{3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its partial transpose has a negative eigenvalue, hence

$$\boxed{\rho_{AB}^{(W)} \text{ is entangled.}}$$

GHZ State

Tracing out qubit 3:

$$\rho_{AB}^{(\text{GHZ})} = \frac{1}{2} (|00\rangle \langle 00| + |11\rangle \langle 11|),$$

which is separable and PPT.

Final Insight

- W state entanglement survives particle loss.
- GHZ entanglement is destroyed by particle loss.

$$\boxed{W \text{ states are robust, while GHZ states are fragile.}}$$

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